KERALA DIPLOMA · REVISION 2026

# Environmental Sustainability and Professional Ethics

Complete handbook for Diploma Engineering students — ecosystem, pollution, ethics, sustainable development, zero waste, green chemistry.

| | | |
|---|---|---|
| PROGRAMME<br>**Diploma Engineering** | SUBJECT CODE<br>**2005** | SEMESTER<br>**4 (as per syllabus)** |
| TEACHING SCHEME<br>**3:0:0:0 (L:T:P:S)** | CREDITS<br>**3** | ESE MARKS<br>**60** |
| CIE MARKS<br>**40** | TOTAL HOURS<br>**45** | |

OFFICIAL

## Source Declaration

This handbook is derived from the official Kerala Polytechnic Revision 2026 syllabus for **Course 2005 — Environmental Sustainability and Professional Ethics**.

⚠️ **Verified inconsistencies in the source PDF:**

- The PDF states "Credits 0" but the teaching scheme is 3:0:0:0, which typically corresponds to 3 credits. This handbook uses 3 credits as the practical value.
- The semester is listed as "Semester 45" — interpreted as Semester 4 (consistent with the course level and typical diploma structure).
- Total instructional hours sum to 45 (12+10+13+10), matching the "45" in the header.

All content has been faithfully expanded for educational purposes while preserving the official syllabus structure and learning outcomes.

AT A GLANCE

## Course Dashboard

**4**
Modules

**45**
Total Hours

**4**
Course Outcomes

**100**
Total Marks

### Course Introduction

**Environmental Sustainability and Professional Ethics** is a course designed for Diploma Engineering students that focuses on protecting the environment and developing ethical values in professional life. It helps students to understand the importance of conserving natural resources, reducing pollution, and promoting sustainable development for present and future generations.

## Pre-requisite

Basic knowledge in environmental science and ethics (Secondary Education level).

## Teaching and Assessment Scheme

| Teaching Scheme / Week | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **L** | **T** | **P** | **S** | **SS/Week** | **Total Hrs/Sem** | **Credits** | **CIE** | **ESE** | **Total** |
| 3 | 0 | 0 | 0 | 4.5 | 45 | 3 | 40 | 60 | 100 |

**L:** Lecture · **T:** Tutorial · **P:** Practical · **S:** Project · **SS:** Self-Learning · **CIE:** Continuous Internal Evaluation · **ESE:** End Semester Examination

GUIDE

# How to Use This Handbook

**1**

### 📖 Understand

Read each module's topics with deep explanations, definitions, and examples.

**2**

### 🖼️ Visualise

Study the SVG diagrams, tables, and concept maps to see relationships.

**3**

### 🧪 Apply

Work through calculators, case studies, and self-learning activities.

**4**

### 📝 Revise

Use Q&A, model paper, revision cards, and glossary for exam preparation.

LEARNING TARGETS

# Course Outcomes

**Course Outcomes (CO) with Bloom's Taxonomy Level**

| CO | Course Outcome | Bloom's Level |
|---|---|---|
| **CO1** | Explain the ecosystem, environmental pollution, important international conventions reg. pollution and legal awareness on environmental policy and laws. | Understand |
| **CO2** | Analyze the significance of environmental ethics and create consciousness about extended ethical approaches to the environment. | Understand |
| **CO3** | Explain sustainable energy resources, their explorations and design a plan for conservation and sustainability in all development programmes. | Understand |
| **CO4** | Explain sustainable engineering principles, zero waste concepts and practices for environmental management. | Apply |

## CO–PO Mapping

**CO-PO Articulation Matrix**

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | - | - | 3 | 3 | - | - | - | - | - |
| CO2 | - | - | 2 | - | 3 | 3 | - | - | - | - | - |
| CO3 | - | - | 2 | - | - | 3 | - | - | - | - | - |
| CO4 | - | - | 3 | - | - | 3 | 2 | - | - | - | - |

Legend: 1 – Low, 2 – Medium, 3 – High, – No Correlation

SYLLABUS

# Curriculum Coverage Map

**Module-wise Coverage**

| Module | Title | Hours | CO | Topics |
|---|---|---|---|---|
| I | Ecosystem & Environment | 12 | CO1 | Environment, Ecosystem, Pollution, International Conventions, Environmental Laws |
| II | Ethics — Professional Ethics — Environmental Ethics | 10 | CO2 | Ethics types, ethical theories, personal/professional ethics, environmental ethics approaches |
| III | Sustainable Development | 13 | CO3 | Man-environment, sustainability, renewable energy, impact analysis, urban planning |
| IV | Ethical Engineering and Environmental Management | 10 | CO4 | Sustainable engineering, zero waste, energy recovery, carbon credit, ISO14000, green chemistry |

**Coverage note:** The official environmental-governance and sustainability topics include the Air Pollution Control Act, Environment Protection Act, National Solar Mission and Green India

Mission. These official names should be retained in examination answers.

CONNECTIONS

## Module Roadmap

**M1**

🌍 **Ecosystem & Environment**

Foundation: environment, pollution, laws, conventions.

**M2**

⚖️ **Ethics**

Moral frameworks: personal, professional, environmental ethics.

**M3**

🌱 **Sustainable Development**

Sustainability, renewable energy, urban planning, policies.

**M4**

♻️ **Ethical Engineering**

Zero waste, carbon credit, ISO14000, green chemistry.

MODULE I

## Ecosystem & Environment

⏱️ 12 Hours | CO1 | Bloom: Understand

This module introduces the fundamental concepts of environmental science, ecosystem structure and function, environmental pollution, major environmental disasters, international conventions, and India's environmental laws.

## ▸Concept and scope of Environmental Science

**Environmental Science** is an interdisciplinary field that integrates physical, biological, and information sciences to study the environment and find solutions to environmental problems. It draws from ecology, chemistry, physics, geology, and social sciences to understand how natural systems function and how human activities affect them.

**Scope of Environmental Science:**

- **Natural Resources:** Study of land, water, air, minerals, and energy resources and their sustainable use.
- **Ecosystem Dynamics:** Understanding energy flow, nutrient cycling, and ecological interactions.
- **Pollution Studies:** Air, water, soil, noise, and radioactive pollution — sources, effects, and control.
- **Environmental Impact Assessment:** Evaluating the environmental consequences of development projects.
- **Environmental Policy and Law:** Legal frameworks for environmental protection.
- **Climate Change:** Causes, impacts, and mitigation strategies.
- **Sustainable Development:** Balancing economic growth with environmental protection.

> **Malayalam support:** [illegible] (Environmental Science) [illegible]

## ▸Environment — Definition and Segments

**Environment** is defined as the sum total of all external conditions and influences that affect the life, development, and survival of an organism. It encompasses all living (biotic) and non-living (abiotic) things surrounding an organism.

**Segments of the Environment:**

- **Atmosphere:** The gaseous envelope surrounding the Earth — composed of nitrogen (78%), oxygen (21%), and trace gases. It protects life from harmful solar radiation and maintains temperature.
- **Lithosphere:** The solid outer layer of the Earth — includes rocks, minerals, and soil. It provides habitat for terrestrial organisms and contains mineral resources.
- **Hydrosphere:** All water bodies — oceans, rivers, lakes, groundwater, and ice caps. Water covers about 71% of the Earth's surface and is essential for all life.
- **Biosphere:** The zone of life on Earth — the sum of all ecosystems where living organisms exist, extending from the deepest ocean trenches to the upper atmosphere.

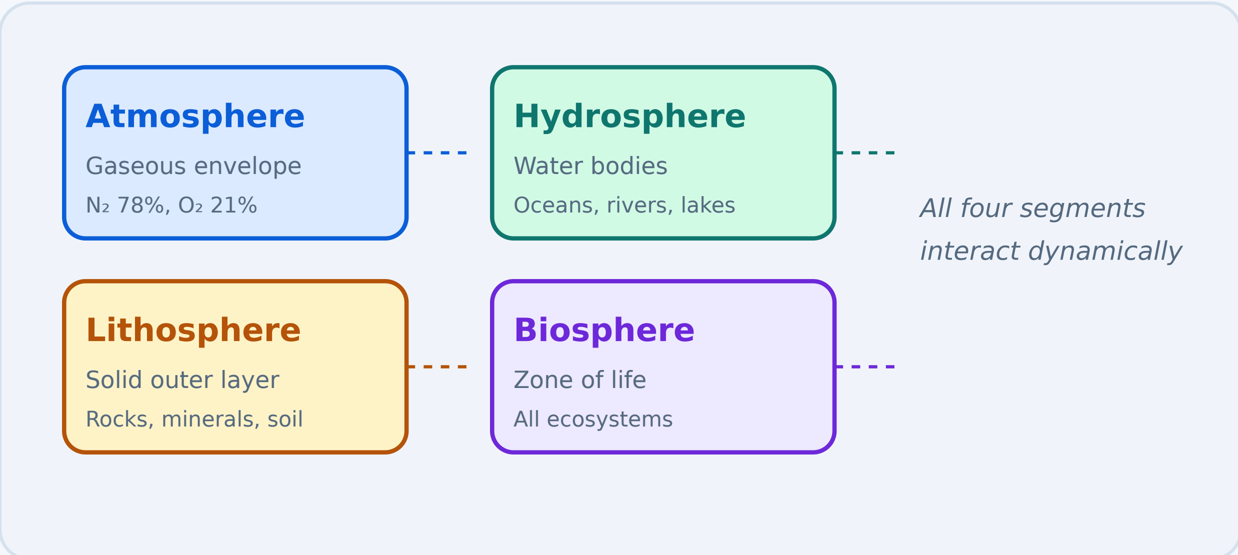


Figure 1: The four segments of the environment — Atmosphere, Lithosphere, Hydrosphere, and Biosphere.

**Malayalam support:** [illegible] (Environment) [illegible] (biotic) [illegible] (abiotic) [illegible].

## ▸Ecosystem — Definition and Components

**Ecosystem** is defined as a community of living organisms (plants, animals, microorganisms) interacting with each other and with their non-living environment (soil, water, air, climate) in a given area. The term was coined by Sir Arthur Tansley in 1935.

**Components of an Ecosystem:**

- **Biotic (Living) Components:**
  - *Producers (Autotrophs):* Green plants, algae, and cyanobacteria that produce food through photosynthesis.
  - *Consumers (Heterotrophs):* Animals that consume producers or other consumers — primary (herbivores), secondary (carnivores), tertiary (top carnivores).
  - *Decomposers (Detritivores):* Bacteria, fungi, and earthworms that break down dead organic matter and recycle nutrients.
- **Abiotic (Non-living) Components:**
  - *Physical Factors:* Sunlight, temperature, rainfall, wind, humidity, soil type, and topography.
  - *Chemical Factors:* Soil pH, mineral content, water salinity, dissolved oxygen, and nutrient availability.

> 💡 **Key insight:** The ecosystem is a functional unit where energy flows and matter cycles. Energy enters as sunlight and flows through trophic levels, while nutrients are recycled through decomposition.

> **Malayalam support:** [illegible] (Ecosystem) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] (producers, consumers, decomposers) [illegible] [illegible] (sunlight, water, soil, temperature) [illegible].

## ▸Energy Flow in an Ecosystem — Food Chain and Food Web

**Food Chain:** A linear sequence of organisms through which nutrients and energy pass as one organism eats another. It shows a single, direct path of energy flow.

*Example:* Grass → Grasshopper → Frog → Snake → Hawk

**Food Web:** A network of interconnected food chains that shows the complex feeding relationships in an ecosystem. It provides a more realistic representation of energy flow.

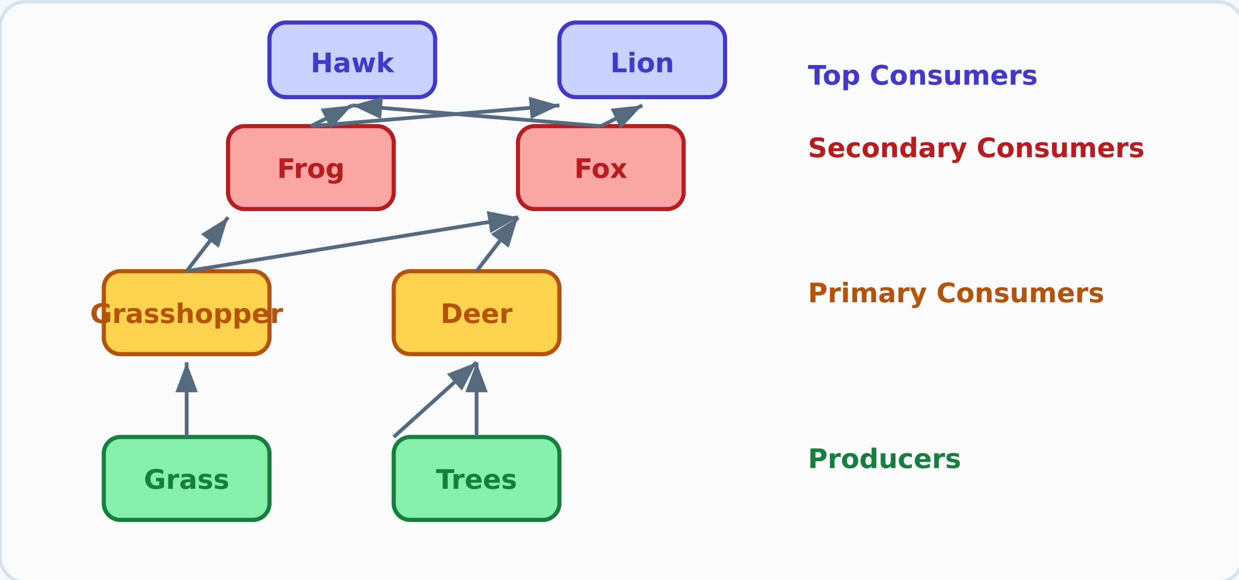


Figure 2: A food web showing the interconnected feeding relationships in an ecosystem.

**Key differences:**

**Food Chain vs Food Web**

| Food Chain | Food Web |
|---|---|
| Single, linear path | Multiple, interconnected paths |
| Simpler representation | More realistic representation |
| One organism at each trophic level | Multiple organisms at each trophic level |
| Energy flow is direct | Energy flow is complex and branched |

> **Malayalam support:** [illegible] (Food chain) [illegible]. [illegible] (Food web) [illegible], [illegible].

## ▸Environmental Pollution — Concepts and Definition

**Environmental pollution** is defined as the introduction of harmful substances or energy (contaminants) into the environment at a rate faster than the environment can naturally disperse, decompose, or recycle them. Pollution can cause damage to living organisms and degrade the quality of air, water, and soil.

**Pollutant:** A substance or form of energy that causes pollution. Pollutants can be natural (volcanic ash, methane) or anthropogenic (industrial emissions, pesticides).

**Classification of Pollutants:**

- **By origin:** Natural (volcanoes, forest fires) vs. Anthropogenic (factories, vehicles)
- **By persistence:** Persistent (take years to degrade — DDT, plastics) vs. Non-persistent (degrade quickly — sewage, food waste)
- **By scale:** Global (climate change, ozone depletion), Regional (acid rain, smog), Local (street-level air pollution, water contamination)
- **By type:** Primary (directly emitted — CO, $SO_2$) vs. Secondary (formed in the environment — ozone, smog)

> 💡 **Remember:** Persistent pollutants like heavy metals and plastics bioaccumulate in food chains, causing long-term ecological damage. Non-persistent pollutants are generally less harmful but can cause acute problems if released in large quantities.

> **Malayalam support:** □□□□□□□□□ □□□□□□□□□ (Environmental pollution) □□□□□□ □□□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□. □□□□□□□□□ (Pollutant) □□□□□□ □□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□.

## ▸Environmental Issues — Ozone Depletion and Global Warming

**Ozone Depletion:**

The ozone layer in the stratosphere (15–50 km above Earth) absorbs 95–99% of the Sun's harmful ultraviolet (UV-B) radiation. Ozone depletion refers to the thinning of this protective layer due to the release of ozone-depleting substances (ODS) like chlorofluorocarbons (CFCs), halons, and methyl bromide.

- **Causes:** CFCs from refrigerants, aerosol propellants, and foam-blowing agents; halons from fire extinguishers; industrial solvents.
- **Effects:** Increased UV-B radiation reaching Earth → skin cancer, cataracts, immune suppression, damage to crops and marine phytoplankton.
- **Solutions:** Montreal Protocol (1987) phased out ODS production; ozone layer is slowly recovering.

**Global Warming:**

Global warming is the long-term increase in Earth's average surface temperature due to the enhanced greenhouse effect. Greenhouse gases (GHGs) — $CO_2$, $CH_4$, $N_2O$, CFCs — trap heat in the atmosphere.

- **Causes:** Burning of fossil fuels, deforestation, industrial processes, agriculture (livestock, rice cultivation), waste decomposition.
- **Effects:** Rising sea levels, extreme weather events, melting glaciers and ice caps, ecosystem shifts, food and water scarcity.
- **Solutions:** Paris Agreement (2015), renewable energy transition, carbon capture, afforestation, energy efficiency.

> ⚠️ **Critical:** The Intergovernmental Panel on Climate Change (IPCC) warns that global warming must be limited to 1.5°C above pre-industrial levels to avoid catastrophic impacts.

> **Malayalam support:** □□□□ □□□□□ (Ozone depletion) □□□□□□□□□□□□□□□□□□ □□□□ □□□□□□□□ □□□ □□□□□□□□ □□□□□□□□□□□□□. □□□□□□□□□ (Global warming) □□□□□□□□ □□□□□□ □□□□□□□□□□ □□□□□□□ □□□□□□□□□□. □□ □□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□ — □□□□□□□□□ □□□□ □□□□ □□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□.

## ▸Case Studies — Major Environmental Disasters

**Bhopal Gas Tragedy (1984):**

- **Location:** Bhopal, Madhya Pradesh, India
- **Incident:** On the night of December 2–3, 1984, methyl isocyanate (MIC) gas leaked from the Union Carbide pesticide plant.
- **Impact:** Over 500,000 people exposed; ~3,800 died immediately; tens of thousands suffered chronic health effects.
- **Cause:** Water entered a storage tank containing MIC, causing a runaway exothermic reaction. Safety systems were non-functional.
- **Lessons:** Industrial safety regulations must be strictly enforced; community emergency preparedness is critical; corporate accountability is essential.

**Minamata Disaster (1956–1960s):**

- **Location:** Minamata Bay, Japan
- **Incident:** Chisso Corporation discharged methylmercury-laden wastewater into the bay, contaminating fish and shellfish.
- **Impact:** Over 2,000 people developed Minamata disease (neurological damage from mercury poisoning); thousands more affected.
- **Cause:** Industrial discharge of mercury used as a catalyst in acetaldehyde production.
- **Lessons:** Industrial waste must be treated before discharge; bioaccumulation of toxins in food chains is a serious threat; government regulation is necessary.

**Chernobyl Nuclear Disaster (1986):**

- **Location:** Chernobyl, Ukraine (then USSR)
- **Incident:** A nuclear reactor explosion released massive amounts of radioactive material into the atmosphere.
- **Impact:** 31 immediate deaths; thousands of cancer cases; large areas rendered uninhabitable; long-term ecological damage.
- **Cause:** A flawed reactor design combined with human error during a safety test.
- **Lessons:** Nuclear safety culture must be paramount; transparent communication is essential during crises; the precautionary principle must guide hazardous technologies.

**Gulf War Oil Spill (1991):**

- **Location:** Persian Gulf, Kuwait
- **Incident:** During the Gulf War, Iraqi forces intentionally released an estimated 6–8 million barrels of oil into the Persian Gulf.
- **Impact:** Severe marine ecosystem damage; destruction of coastal habitats; massive economic losses to fishing and tourism.
- **Cause:** Deliberate release of oil as a tactic to hinder US forces and create an ecological disaster.
- **Lessons:** Warfare can cause massive environmental damage; international laws must protect the environment during armed conflict; oil spill response protocols need strengthening.

🛡️ **Engineering takeaway:** These disasters highlight the critical importance of safety engineering, regulatory compliance, risk assessment, and corporate social responsibility in all industries.

**Malayalam support:** [illegible] [illegible] (1984), [illegible] [illegible] (1956), [illegible] [illegible] (1986), [illegible] [illegible] [illegible] (1991) [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible], [illegible] [illegible], [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible].

## ▸Important International Conventions and Conferences

**Major International Environmental Conventions and Conferences**

| Convention/Conference | Year | Key Focus |
|---|---|---|
| **Stockholm Conference** | 1972 | First UN conference on the human environment; established UNEP; declared that environmental protection is a human right. |
| **Basel Convention** | 1989 | Controls the transboundary movement of hazardous wastes and their disposal; aims to reduce hazardous waste generation. |
| **Montreal Protocol** | 1987 | Phases out ozone-depleting substances (CFCs, halons, etc.); widely considered the most successful environmental treaty. |
| **Kyoto Protocol** | 1997 | Legally binding emissions reduction targets for developed countries (Annex I); established carbon trading mechanisms. |
| **Paris Agreement** | 2015 | Global framework to limit warming to well below 2°C, with efforts to limit to 1.5°C; nationally determined contributions (NDCs). |
| **Earth Summit (UNCED)** | 1992 | Rio de Janeiro; adopted Agenda 21, the Rio Declaration, and conventions on climate change and biodiversity. |
| **Agenda 21** | 1992 | Non-binding action plan for sustainable development at global, national, and local levels. |
| **Copenhagen Summit (COP15)** | 2009 | Attempted to negotiate a successor to Kyoto; produced the Copenhagen Accord with a goal of limiting warming to 2°C. |
| **MDG (Millennium Development Goals)** | 2000 | 8 goals including poverty reduction, education, and environmental sustainability; replaced by SDGs. |
| **SDG (Sustainable Development Goals)** | 2015 | 17 goals with 169 targets for 2030, including climate action, clean energy, and sustainable cities. |

💡 **Note:** The Montreal Protocol is often cited as the most successful international environmental agreement — it phased out 99% of ozone-depleting substances and the ozone layer is now recovering.

**Malayalam support:** [illegible] (1972), [illegible] (1989), [illegible] (1987), [illegible] (1997), [illegible] (2015) [illegible].

## ▸Environmental Policy and Laws of India

**Major Environmental Laws of India**

| Law | Year | Key Provisions |
|---|---|---|
| **Indian Forest Act** | 1927 | Regulates the use and management of forests; provides for the reservation and protection of forests; controls forest produce. |
| **Water (Prevention and Control of Pollution) Act** | 1974 | Prevents and controls water pollution; establishes Central and State Pollution Control Boards; regulates discharge of pollutants into water bodies. |
| **Air (Prevention and Control of Pollution) Act** | 1981 | Prevents and controls air pollution; empowers Pollution Control Boards to set emission standards; regulates industries. |
| **Environment (Protection) Act (EPA)** | 1986 | Umbrella legislation for environmental protection; empowers the Central Government to take measures to protect the environment; sets standards for emissions and effluents. |

**Key features of the EPA 1986:**

- Provides for the protection and improvement of environmental quality.
- Empowers the Central Government to coordinate actions and set standards.
- Requires environmental impact assessment (EIA) for certain projects.
- Provides for penalties for non-compliance — imprisonment up to 5 years and fines.

> 🛡️ **Key takeaway:** The EPA 1986 is the most comprehensive environmental law in India, serving as the backbone for environmental governance and enforcement.

> **Malayalam support:** [illegible]: [illegible] (1927), [illegible] (1974), [illegible] (1981), [illegible] (1986). [illegible].

> 📌 **Module I Summary:** This module covered the fundamentals of environmental science — the definition and segments of environment, ecosystem components and energy flow, pollution types and classification, major environmental issues (ozone depletion, global warming), four major case studies, key international conventions, and India's environmental laws. Understanding these concepts is essential for the modules that follow.

MODULE II

# Ethics — Professional Ethics — Environmental Ethics

⏱️ 10 Hours | CO2 | Bloom: Understand

This module explores the concept of ethics, its types, ethical theories, personal and professional ethics, and specifically focuses on environmental ethics — its historical development and different ethical approaches to the environment.

### ▸Ethics — Term, Etymology, Nature and Scope

**Ethics** (from the Greek *ethos*, meaning "character" or "custom") is the branch of philosophy that deals with moral principles, values, and the rules of conduct that govern a person's behavior or the conducting of an activity. It is concerned with what is good and bad, right and wrong.

**Nature of Ethics:**

- **Normative:** Ethics prescribes how people ought to behave.
- **Descriptive:** Ethics describes how people actually behave.
- **Philosophical:** Ethics involves critical reasoning about moral issues.
- **Practical:** Ethics has real-world applications in all spheres of human life.

**Scope of Ethics:**

- **Personal Ethics:** Individual moral beliefs and values.
- **Social Ethics:** Moral norms within groups and communities.
- **Professional Ethics:** Moral principles specific to professions.
- **Environmental Ethics:** Moral considerations regarding the environment.
- **Business Ethics:** Ethical principles in commerce and industry.

> **Malayalam support:** [illegible] (Ethics) [illegible], [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible].

## ▸Types of Ethics — Normative, Meta, Applied

**Types of Ethics**

| Type | Definition | Key Question |
|---|---|---|
| **Normative Ethics** | Studies what is morally right and wrong; sets standards for conduct. | "What should I do?" |
| **Meta-Ethics** | Examines the nature of moral judgment, language, and concepts. | "What does 'good' mean?" |
| **Applied Ethics** | Applies ethical principles to specific real-world situations. | "What should we do in this specific case?" |

**Normative Ethics** includes consequentialism (focus on outcomes), deontology (focus on duties/rules), and virtue ethics (focus on character).

**Meta-Ethics** explores the meaning of moral terms like "good," "right," "justice," and whether moral truths are objective or subjective.

**Applied Ethics** includes bioethics, business ethics, environmental ethics, legal ethics, and medical ethics.

> 💡 **Connection:** Environmental ethics is a branch of applied ethics — it applies moral principles to environmental issues and human-nature relationships.

> **Malayalam support:** [illegible] (Ethics) [illegible]: [illegible] ([illegible]), [illegible]-[illegible] ([illegible]), [illegible] ([illegible]).

## ▸Ethical Theories — Deontology, Consequentialism, Virtue Ethics

**Major Ethical Theories**

| Theory | Key Proponent | Core Idea | Focus |
|---|---|---|---|
| **Deontology** | Immanuel Kant | Actions are morally right if they follow universal moral rules/duties. | Duty and rules |
| **Consequentialism** | Jeremy Bentham, John Stuart Mill | The morality of an action depends on its outcomes/consequences. | Results and outcomes |
| **Virtue Ethics** | Aristotle | Focuses on the character of the moral agent rather than rules or outcomes. | Character and virtues |

**Deontology** — "Act only according to that maxim whereby you can at the same time will that it should become a universal law." (Kant's Categorical Imperative). Actions are judged by their adherence to moral duties, regardless of consequences.

**Consequentialism** — "The greatest good for the greatest number." (Utilitarianism). Actions are judged by their outcomes — the right action is the one that produces the best overall consequences.

**Virtue Ethics** — Focuses on the development of moral character traits (virtues) — honesty, courage, compassion, integrity. A virtuous person naturally acts in morally right ways.

> 💡 **Application:** In environmental ethics, deontology might argue that we have a duty to protect the environment regardless of consequences; consequentialism might argue for environmental protection if it leads to the best overall outcomes; virtue ethics might argue that a virtuous person would care for the environment as part of good character.

> **Malayalam support:** □□□□□□□□□□□□ (Deontology) — □□□□□□□ □□□□□□□□□□□ □□□□□□□□; □□□□□□□□□□□□□□□□□ (Consequentialism) — □□□□□□□□□ □□□□□□□□□□□□ □□□□□□□□; □□□□□□□□ □□□□□□□□□□ (Virtue ethics) — □□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□.

## ▸Personal and Professional Ethics

**Personal Ethics vs Professional Ethics**

| Personal Ethics | Professional Ethics |
|---|---|
| Based on individual moral values and beliefs. | Based on the standards and codes of a profession. |
| Developed through upbringing, religion, culture, and personal reflection. | Developed through professional training, codes of conduct, and peer review. |
| Applied in personal life — relationships, family, personal decisions. | Applied in professional life — workplace, clients, colleagues, public. |
| Examples: Honesty, kindness, loyalty, personal integrity. | Examples: Confidentiality, competence, impartiality, conflict of interest management. |

**Moral standpoints of personal ethics:** Personal ethics are shaped by individual conscience, religious beliefs, cultural norms, and personal experiences. They form the foundation of a person's moral identity.

**Moral standpoints of professional ethics:** Professional ethics are codified standards that guide the conduct of professionals. They are designed to protect the public, maintain trust, and ensure professional accountability.

> 🛡️ **Key principle:** A professional must integrate personal ethical values with professional ethical standards. Conflicts between personal and professional ethics should be resolved through careful moral reasoning and, if necessary, consultation.

> **Malayalam support:** [illegible] (Personal ethics) [illegible]. [illegible] (Professional ethics) [illegible].

## ▸Types of Professional Ethics

**Types of Professional Ethics**

| Type | Definition | Key Principles |
|---|---|---|
| **Business Ethics** | Moral principles applied to business and commerce. | Fairness, transparency, honesty, corporate social responsibility. |
| **Legal Ethics** | Ethical standards for lawyers and legal professionals. | Confidentiality, loyalty to client, impartiality, professional competence. |
| **Medical Ethics** | Ethical principles for healthcare professionals. | Patient autonomy, beneficence, non-maleficence, confidentiality. |
| **Environmental Ethics** | Moral principles concerning the environment and human-nature relationships. | Stewardship, sustainability, intrinsic value of nature, intergenerational equity. |

> **Malayalam support:** [illegible] (Business ethics), [illegible] (Legal ethics), [illegible] (Medical ethics), [illegible] (Environmental ethics) [illegible].

## ▸Environmental Ethics — Importance and Historical Development

**Environmental ethics** is the branch of ethics that extends moral consideration to the natural environment — including non-human animals, plants, ecosystems, and the planet as a whole.

**Importance:**

- Provides a moral framework for addressing environmental degradation and climate change.
- Challenges the assumption that nature is merely a resource for human use.
- Recognizes the intrinsic value of nature — nature has worth beyond its utility to humans.
- Promotes intergenerational justice — protecting the environment for future generations.
- Guides environmental policy, sustainable development, and conservation efforts.

**Historical Development:**

- **Early philosophical foundations:** Ancient Greek and Eastern philosophies (Stoicism, Buddhism, Jainism) recognized respect for nature.
- **19th Century:** Transcendentalists (Thoreau, Emerson) — nature as a source of spiritual and moral inspiration.
- **1960s-70s:** Modern environmental ethics emerges — Rachel Carson's *Silent Spring* (1962), the first Earth Day (1970), and the founding of environmental philosophy as a discipline.
- **Key thinkers:** Aldo Leopold (land ethic — "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community."), Arne Naess (deep ecology), Peter Singer (animal ethics).
- **Contemporary:** Environmental ethics now addresses climate justice, biodiversity loss, sustainable development, and the rights of nature.

> 💡 **Key insight:** The historical development of environmental ethics reflects a growing awareness that human well-being is inextricably linked to the health of the natural world.

> **Malayalam support:** [illegible] [illegible] (Environmental ethics) [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] '[illegible] [illegible]' [illegible] [illegible] [illegible] [illegible].

## ▸Ethical Approaches to the Environment — Anthropocentrism, Biocentrism, Ecocentrism

**Ethical Approaches to the Environment**

| Approach | Definition | Key Idea | Focus |
|---|---|---|---|
| **Anthropocentrism** | Human-centered ethics — only humans have intrinsic value. | Nature is a resource for human use; environmental protection is justified only if it benefits humans. | Human interests |
| **Biocentrism** | Life-centered ethics — all living beings have intrinsic value. | Every living organism (plants, animals, microorganisms) has moral standing; humans are not superior to other life forms. | All living beings |
| **Ecocentrism** | Ecosystem-centered ethics — the whole ecosystem has intrinsic value. | Ecosystems, species, and ecological processes have moral worth; the health of the whole system is paramount. | Entire ecosystems |

**Anthropocentrism** is the dominant view in mainstream economics and policy — it values nature primarily for its utility to humans. This approach can lead to overexploitation of natural resources.

**Biocentrism** extends moral consideration to all living beings. Key proponents include Albert Schweitzer (reverence for life) and Paul Taylor (respect for nature). This approach recognizes that all life has a good of its own.

**Ecocentrism** is the broadest approach — it values ecosystems as wholes, not just individual organisms. Key proponents include Aldo Leopold (land ethic) and Arne Naess (deep ecology). This approach emphasizes ecological interconnectedness and systemic health.

> 💡 **Evolution:** Environmental ethics has evolved from anthropocentrism (human-focused) → biocentrism (life-focused) → ecocentrism (ecosystem-focused). Each approach broadens the circle of moral consideration.

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📌 **Module II Summary:** This module explored the foundations of ethics — its definition, types (normative, meta, applied), and major ethical theories (deontology, consequentialism, virtue ethics). It distinguished personal ethics from professional ethics and examined professional ethics in business, law, medicine, and environment. The module focused on environmental ethics — its historical development and three key approaches: anthropocentrism, biocentrism, and ecocentrism.

MODULE III

## Sustainable Development

⏱️ 13 Hours | CO3 | Bloom: Understand

This module examines the relationship between man and environment, the concept of sustainability, renewable energy sources, environmental impact assessment, sustainable urban planning, and India's sustainable development initiatives.

## ▸Man-Environment Relationship and Anthropogenic Effects

**Man-Environment Relationship:** Humans are both part of and dependent on the environment. The relationship has evolved over time — from early hunter-gatherer societies that lived in harmony with nature, to agricultural societies that modified landscapes, to industrial societies that have significantly altered the Earth's ecosystems.

**Anthropogenic Effects (Human-induced impacts):**

- **Pollution:** Air, water, soil, and noise pollution from industrial, agricultural, and domestic activities. Toxic chemicals, plastics, and microplastics contaminate ecosystems.
- **Deforestation:** Clearing of forests for agriculture, urbanization, and timber — leading to habitat loss, soil erosion, and reduced carbon sequestration.
- **Soil Erosion:** Loss of topsoil due to deforestation, overgrazing, unsustainable farming, and construction — reducing agricultural productivity.
- **Mass Extinction:** The current rate of species extinction is 1,000–10,000 times higher than natural background rates. Habitat loss, climate change, pollution, and invasive species are driving the sixth mass extinction.
- **Climate Change:** Greenhouse gas emissions from burning fossil fuels, deforestation, and agriculture are causing global warming and climate disruption.

> ⚠️ **Critical:** The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) reports that 1 million species are threatened with extinction due to human activities.

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## ▸Concept of Sustainability and Need for Sustainability

**Sustainability** is the ability to meet the needs of the present without compromising the ability of future generations to meet their own needs. This definition was popularized by the Brundtland Commission in 1987.

**Three Pillars of Sustainability:**

- **Environmental Sustainability:** Protecting and preserving natural resources, ecosystems, and biodiversity.
- **Social Sustainability:** Ensuring equity, social justice, community well-being, and cultural preservation.
- **Economic Sustainability:** Creating economic systems that are viable, efficient, and equitable over the long term.

**Need for Sustainability:**

- **Resource depletion:** Finite natural resources (fossil fuels, minerals, fresh water) are being consumed faster than they can be replenished.
- **Environmental degradation:** Pollution, climate change, and habitat loss are threatening ecosystem services that support human life.
- **Population growth:** The global population is projected to reach 9.7 billion by 2050, increasing resource demands.
- **Intergenerational justice:** We have a moral obligation to leave a habitable planet for future generations.
- **Economic stability:** Unsustainable practices lead to economic instability — resource shortages, price volatility, and systemic risks.

> 💡 **Key insight:** Sustainability is not just an environmental issue — it is an economic, social, and ethical imperative. The three pillars must be balanced for true sustainability.

> **Malayalam support:** [illegible] (Sustainability) [illegible]. [illegible]: [illegible], [illegible], [illegible].

## ▸Sustainable Development — Approaches and Strategies

**Sustainable development** is the process of achieving economic growth and social progress while protecting and enhancing the environment for future generations.

**Approaches:**

- **Top-down approach:** Government-led policies, regulations, and international agreements.
- **Bottom-up approach:** Community-led initiatives, local solutions, and grassroots movements.
- **Market-based approach:** Using economic incentives (carbon pricing, green bonds, eco-labeling) to drive sustainable behavior.
- **Integrated approach:** Combining top-down, bottom-up, and market-based strategies for holistic sustainability.

**Strategies for Sustainable Development:**

- **Green Economy:** An economic system that is low-carbon, resource-efficient, and socially inclusive.
- **Circular Economy:** An economic model that eliminates waste through reuse, repair, remanufacturing, and recycling.
- **Decarbonization:** Transitioning from fossil fuels to renewable energy sources to reduce greenhouse gas emissions.
- **Eco-innovation:** Developing products, processes, and services that reduce environmental impact.
- **Education and Awareness:** Empowering citizens with knowledge and skills for sustainable living.
- **Policy Integration:** Embedding sustainability into all policy areas — planning, finance, industry, agriculture, transportation.

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## ▸Environment Friendly Products and Technologies

**Environment-friendly products** (green products) are products that have a reduced environmental impact throughout their life cycle — from raw material extraction to production, use, and disposal.

**Characteristics:**

- Made from renewable or recycled materials.
- Energy-efficient in production and use.
- Non-toxic and biodegradable.
- Minimal packaging or recycled packaging.
- Durable and repairable (designed for longevity).

**Environment-friendly technologies (green technologies):**

- **Renewable energy technologies:** Solar panels, wind turbines, hydroelectric systems, biomass energy.
- **Energy efficiency technologies:** LED lighting, high-efficiency motors, smart grid systems, energy-efficient appliances.
- **Water conservation technologies:** Drip irrigation, rainwater harvesting, greywater recycling, desalination.
- **Waste management technologies:** Composting, anaerobic digestion, recycling technologies, waste-to-energy.
- **Pollution control technologies:** Scrubbers, electrostatic precipitators, catalytic converters, membrane bioreactors.
- **Green building technologies:** Insulation materials, solar heating, green roofs, smart windows.

> 💡 **Note:** Green technologies are not just environmentally beneficial — they also create economic opportunities, reduce costs, and enhance energy security.

> **Malayalam support:** [illegible] [illegible] [illegible] [illegible] (Green products and technologies) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

## ▸Equity in Resource Distribution and Consumption

**Equity in resource distribution** refers to the fair allocation of natural resources among individuals, communities, and nations. It addresses the fact that resource consumption is highly unequal — the richest 20% of the global population consumes 80% of the world's resources.

**Key issues:**

- **Resource inequality:** Wealthy nations consume disproportionately more resources per capita than developing nations.
- **Ecological footprint:** The average ecological footprint of a person in the US is 8.2 global hectares, compared to 0.8 in India.
- **Climate injustice:** Developing countries are most vulnerable to climate change but have contributed the least to greenhouse gas emissions.
- **Intergenerational equity:** Current generations are consuming resources that rightfully belong to future generations.

**Principles of equity in resource distribution:**

- **Equal rights:** All people have an equal right to access and benefit from natural resources.
- **Need-based allocation:** Resources should be allocated based on needs, not just purchasing power.
- **Common but differentiated responsibilities:** All nations have a responsibility to protect the environment, but developed nations have greater responsibility due to their historical emissions.
- **Polluter pays principle:** Those who pollute should bear the costs of environmental damage.

> 🛡️ **Ethical principle:** Environmental justice requires that the benefits and burdens of environmental policies are distributed fairly among all communities.

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## ▸Renewable Energy Sources and Sustainable Technologies

**Overview of Renewable Energy Sources**

| Source | Energy Conversion | Advantages | Challenges |
|---|---|---|---|
| **Solar** | Solar → Electricity (PV) or Heat (thermal) | Abundant, scalable, low maintenance | Intermittent, land use, storage |
| **Wind** | Kinetic → Electricity | Low operating cost, scalable | Intermittent, noise, visual impact |
| **Hydro** | Potential/Kinetic → Electricity | Reliable, dispatchable, long life | Ecological impact, displacement |
| **Biomass** | Chemical → Heat/Electricity | Waste utilization, dispatchable | Emissions, land competition |
| **Geothermal** | Thermal → Electricity/Heat | Baseload, low emissions | Location-specific, cost |

### Sustainable technologies in energy production and consumption:

- **Solar photovoltaics (PV):** Convert sunlight directly into electricity; efficiency is improving (25%+ for commercial modules).
- **Wind turbines:** Onshore and offshore wind farms; capacity factor of 35–55%.
- **Lithium-ion batteries:** Energy storage for grid and electric vehicles; costs have dropped 90% in the last decade.
- **Smart grids:** Use digital technology to manage electricity supply and demand efficiently.
- **Green hydrogen:** Hydrogen produced from renewable electricity via electrolysis; zero-carbon fuel for transport and industry.

> 💡 **Key trend:** The cost of renewable energy has fallen dramatically — solar is now the cheapest source of electricity in most parts of the world.

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## ▸Environmental Impact Analysis and Environmental Audit

**Environmental Impact Assessment (EIA):**

EIA is a systematic process for evaluating the likely environmental consequences of a proposed project or development. It aims to ensure that environmental factors are considered in decision-making.

**Key steps in the EIA process:**

- **Screening:** Determining whether a project requires a full EIA.
- **Scoping:** Identifying the key environmental issues to be assessed.
- **Impact analysis:** Predicting and evaluating the environmental impacts of the proposed project.
- **Mitigation:** Proposing measures to avoid, reduce, or compensate for adverse impacts.
- **Environmental Management Plan (EMP):** Outlining how impacts will be managed during construction and operation.
- **Public consultation:** Engaging with affected communities and stakeholders.
- **Decision-making:** Using the EIA report to inform project approval.
- **Monitoring:** Tracking environmental performance during project implementation.

**Environmental Audit:**

An environmental audit is a systematic, documented, periodic, and objective review of an organization's activities, processes, and systems to assess their environmental performance and compliance with regulatory requirements.

**Purpose and benefits of Environmental Audit:**

- **Compliance verification:** Ensuring compliance with environmental laws and regulations.
- **Performance improvement:** Identifying opportunities for reducing environmental impact.
- **Risk management:** Identifying and mitigating environmental risks.
- **Cost savings:** Reducing waste, energy consumption, and resource use.
- **Corporate reputation:** Demonstrating environmental responsibility to stakeholders.
- **Certification:** Supporting ISO 14001 certification and other environmental management standards.

> 💡 **Difference:** EIA is conducted before a project begins (prospective), while environmental audits are conducted during or after operations (retrospective).

> **Malayalam support:** [illegible] (EIA) [illegible]. [illegible] (Environmental audit) [illegible].

## ▸Sustainable Development Initiatives in India

**Major Sustainable Development Initiatives in India**

| Initiative | Year | Key Focus |
|---|---|---|
| **National River Conservation Plan** | 1995 | Cleanup and conservation of rivers, including the Ganga Action Plan. |
| **Smart City Projects** | 2015 | Developing 100 smart cities with sustainable infrastructure and digital technology. |
| **Swachh Bharat Abhiyan (Clean India Mission)** | 2014 | Eliminating open defecation, improving waste management, and promoting cleanliness. |
| **National Clean Energy Fund** | 2010 | Investing in clean energy projects (renewable energy, energy efficiency). |
| **National Water Mission** | 2008 | Integrated water resource management, water conservation, and reducing water stress. |

### Emerging trends and future directions in environmental sustainability:

- **Green hydrogen:** India's National Hydrogen Mission aims to make India a global hub for green hydrogen production.
- **Circular economy:** Transitioning from linear "take-make-dispose" to circular models that eliminate waste.
- **Nature-based solutions:** Using natural ecosystems (forests, wetlands, mangroves) for climate adaptation and mitigation.
- **Digital sustainability:** Using AI, IoT, and big data for environmental monitoring and management.
- **Climate finance:** Scaling up investment in climate adaptation and mitigation projects.

> **Malayalam support:** [illegible]

## ▸Urbanization and Sustainable Urban Planning

**Urbanization** is the process by which an increasing proportion of a population lives in cities and towns. Over 55% of the global population now lives in urban areas, projected to reach 68% by 2050.

**Environmental impacts of urbanization:**

- **Air pollution:** Emissions from vehicles, industry, and construction.
- **Water pollution:** Urban runoff, sewage discharge, industrial effluents.
- **Heat island effect:** Urban areas are significantly warmer than rural areas due to asphalt, concrete, and reduced vegetation.
- **Biodiversity loss:** Habitat fragmentation and destruction due to urban expansion.
- **Resource consumption:** High demand for water, energy, and materials.
- **Waste generation:** Municipal solid waste is a major urban challenge.

**Sustainable urban planning and green infrastructure:**

- **Green buildings:** Energy-efficient, water-efficient, using sustainable materials.
- **Urban parks and green spaces:** Parks, gardens, green roofs, and urban forests that provide cooling, recreation, and habitat.
- **Sustainable transportation:** Public transit, cycling infrastructure, pedestrian-friendly design, electric vehicles.
- **Integrated urban water management:** Rainwater harvesting, stormwater management, water-sensitive urban design.
- **Waste management:** Segregation, recycling, composting, and waste-to-energy.
- **Compact city design:** Walkable neighborhoods, mixed-use development, and reduced urban sprawl.

> 💡 **Key principle:** Sustainable urban planning integrates environmental, economic, and social considerations to create livable, resilient, and equitable cities.

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> 📌 **Module III Summary:** This module covered the man-environment relationship, anthropogenic impacts, the concept of sustainability, sustainable development approaches, green products and technologies, equity in resource distribution, renewable energy sources, EIA and environmental audit, India's sustainable development initiatives, urbanization impacts, and sustainable urban planning strategies.

MODULE IV

# Ethical Engineering and Environmental Management

⏱️ 10 Hours | CO4 | Bloom: Apply

This module introduces sustainable engineering principles, zero waste concepts and practices, energy recovery from waste, carbon credit and footprint, environmental management in industry, ISO14000, biotechnology, and green chemistry for sustainable development.

## ▸Sustainable Engineering Principles

**Sustainable engineering** is the design, development, and operation of systems and processes that meet human needs while preserving the environment and natural resources for future generations.

**Scope of sustainable engineering:**

- Designing products and processes with minimal environmental impact.
- Using renewable and recyclable materials.
- Reducing energy and water consumption.
- Minimizing waste and emissions.
- Considering the full life cycle of products.

**Triple Bottom Line:**

- **Economic sustainability:** Projects must be economically viable and profitable.
- **Social sustainability:** Projects must benefit communities, ensure equity, and respect human rights.
- **Environmental sustainability:** Projects must protect and enhance the environment.

**Life Cycle Analysis (LCA):**

LCA is a methodology for assessing the environmental impacts of a product, process, or service throughout its entire life cycle — from raw material extraction to disposal (cradle-to-grave).

**Stages of LCA:**

- **Goal and scope definition:** Defining the purpose and boundaries of the assessment.
- **Inventory analysis:** Compiling all inputs (materials, energy) and outputs (emissions, waste) for each life cycle stage.
- **Impact assessment:** Evaluating the environmental impacts of the inputs and outputs.
- **Interpretation:** Drawing conclusions and making recommendations.

> 💡 **Note:** LCA helps engineers make informed decisions about material selection, process design, and product development to reduce environmental impact.

> **Malayalam support:** [illegible] [illegible] (Sustainable engineering) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] (Triple bottom line) [illegible], [illegible], [illegible] [illegible] [illegible] [illegible] [illegible].

## ▸Zero Waste Concepts and Practices

**Zero waste** is a philosophy and design principle that aims to eliminate waste — not just manage it. The goal is to send nothing to landfill or incineration by designing products and systems that are circular.

**Zero waste principles:**

- **Refuse:** Avoid unnecessary items and packaging.
- **Reduce:** Minimize consumption and waste generation.
- **Reuse:** Use items multiple times; repair and repurpose.
- **Recycle:** Convert waste materials into new products.
- **Rot:** Compost organic waste.
- **Rethink:** Redesign systems and products for circularity.

**3R Principle (Reduce, Reuse, Recycle):**

- **Reduce:** Using fewer resources and generating less waste. Examples: minimalist packaging, digital documents, efficient design.
- **Reuse:** Using items multiple times instead of disposing after single use. Examples: refillable bottles, reusable shopping bags, repairs.
- **Recycle:** Processing waste materials into new products. Examples: paper recycling, plastic recycling, metal recycling.

**Need for recycling:**

- Conserves natural resources (saving virgin materials).
- Reduces energy consumption (recycling uses less energy than producing from raw materials).
- Reduces greenhouse gas emissions.
- Reduces landfill use and pollution.
- Creates economic opportunities and jobs.

**Case studies of successful zero waste initiatives:**

- **Kamaraj, Japan:** A small town that has achieved 80%+ recycling rates through rigorous waste segregation and community engagement.
- **San Francisco, USA:** Achieved 80% waste diversion through mandatory composting and recycling.
- **Zero Waste Kerala:** Various initiatives in Kerala for decentralized waste management, biogas plants, and recycling.

> ⚠️ **Challenge:** Zero waste requires systemic change — from product design to consumer behavior to waste management infrastructure.

> **Malayalam support:** □□□□ □□□□□□□□ (Zero waste) □□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□ □□□□□□□□□□□□□□□□□□□□□. 3R — Reduce, Reuse, Recycle — □□□□□□□ □□□□□□□□ □□□□□□□□□□□□.

## ▸Energy Recovery from Waste

**Energy Recovery Methods from Waste**

| Type | Method | Description | Products |
|---|---|---|---|
| **Biochemical** | **Landfill gas recovery** | Capturing methane produced by decomposition of organic waste in landfills. | Methane (biogas) → electricity/heat |
| | **Composting** | Aerobic decomposition of organic waste to produce compost and $CO_2$. | Compost (fertilizer), heat |
| **Thermochemical** | **Incineration** | Burning of waste at high temperatures (800–1000°C) to generate heat/electricity. | Electricity, heat, ash |
| | **Gasification** | Partial oxidation of waste at high temperature (700–1400°C) to produce syngas. | Syngas ($CO + H_2$) → electricity/chemicals |
| | **Pyrolysis** | Thermal decomposition of waste in the absence of oxygen (400–700°C). | Bio-oil, syngas, char |

**Biochemical methods:**

- **Landfill gas recovery:** Methane from landfills can be captured and used for electricity generation or as a fuel. Reduces greenhouse gas emissions compared to venting methane.
- **Composting:** Organic waste (food, garden waste) decomposes aerobically to produce nutrient-rich compost for agriculture.

**Thermochemical methods:**

- **Incineration:** Most common waste-to-energy method; reduces waste volume by 80–90%; produces electricity and heat.
- **Gasification:** Produces syngas ($CO + H_2$) which can be used for electricity, heat, or chemical production; lower emissions than incineration.
- **Pyrolysis:** Produces bio-oil (can be upgraded to fuel), syngas, and char (can be used as soil amendment).

💡 **Note:** Energy recovery from waste is a key strategy in the waste hierarchy — it is preferred over landfill but less preferred than prevention, reuse, and recycling.

**Malayalam support:** [illegible] [illegible] [illegible] [illegible] (Energy recovery from waste) [illegible] [illegible] [illegible]: [illegible] (landfill gas, composting), [illegible] (incineration, gasification, pyrolysis).

## ▸Carbon Credit and Carbon Footprint

**Carbon Credit:**

A carbon credit is a tradable permit that represents one metric ton of carbon dioxide (or its equivalent in other greenhouse gases) that has been reduced, avoided, or removed from the atmosphere. Carbon credits are used in carbon trading markets to incentivize emissions reductions.

**How carbon credits work:**

- Companies or countries are assigned emissions caps.
- If they emit less than their cap, they earn carbon credits.
- These credits can be sold to other entities that exceed their caps.
- This creates a market incentive to reduce emissions.

**Types of carbon credits:**

- **Compliance credits:** Used under regulatory schemes (e.g., EU ETS, Kyoto Protocol).
- **Voluntary credits:** Used by organizations voluntarily to offset their emissions.

**Carbon Footprint:**

A carbon footprint is the total amount of greenhouse gases (GHGs) emitted directly or indirectly by an individual, organization, product, or activity — measured in units of carbon dioxide equivalent ($CO_2e$).

**Components of a carbon footprint:**

- **Direct emissions (Scope 1):** Emissions from sources owned or controlled by the entity (e.g., company vehicles, on-site fuel combustion).
- **Indirect emissions (Scope 2):** Emissions from purchased electricity, heat, or steam.
- **Other indirect emissions (Scope 3):** Emissions from the entire value chain — supply chain, product use, waste disposal.

$$\text{Carbon Footprint } (CO_2e) = \Sigma\,(\text{Activity Data} \times \text{Emission Factor})$$

Where: Activity Data = quantity of the activity (e.g., kWh of electricity, litres of fuel), Emission Factor = $CO_2e$ per unit of activity.

💡 **Key insight:** Carbon footprint accounting is the first step in emissions reduction — what gets measured gets managed.

**Malayalam support:** □□□□□ □□□□□□□□□□□ (Carbon credit) □□□ □□ $CO_2$ □□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□. □□□□□ □□□□□□□□□□□□□□□ (Carbon footprint) □□□ □□□□□□□□□ □□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□□ □□□□□□ □□□□□□□ □□□□□□□□□□□□.

## ▸Environmental Management in the Fabrication Industry

The fabrication industry (metal working, manufacturing, machining) has significant environmental impacts — energy consumption, waste generation (metal scrap, cutting fluids, sludge), air emissions (VOCs, particulates), and water pollution.

**Key environmental issues in fabrication:**

- **Energy consumption:** High energy use for welding, machining, heat treatment.
- **Waste generation:** Metal scrap, used cutting fluids, abrasive materials, packaging waste.
- **Air emissions:** Welding fumes, volatile organic compounds (VOCs) from paints and solvents, particulate matter.
- **Water pollution:** Discharge of contaminated water from cleaning and rinsing operations.
- **Noise pollution:** High noise levels from machinery and processes.

**Environmental management strategies:**

- **Energy efficiency:** Using energy-efficient motors, LED lighting, and process optimization.
- **Waste minimization:** Recycling metal scrap, reducing material use, reusing cutting fluids.
- **Emission control:** Using fume extraction systems, low-VOC paints, and dust collection.
- **Water conservation:** Water recycling, efficient cleaning systems, and reducing water use.
- **Green procurement:** Using sustainable materials and suppliers with environmental certifications.
- **Environmental Management System (EMS):** Implementing ISO 14001 or similar standards.

> **Malayalam support:** [illegible] (Environmental management) [illegible], [illegible], [illegible], [illegible].

## ▸ISO 14000 — Implementation and Benefits

**ISO 14000** is a family of international standards for environmental management systems (EMS). The most well-known standard is ISO 14001, which specifies requirements for an EMS.

**ISO 14001 key requirements:**

- **Environmental policy:** A documented policy with commitment to compliance and continuous improvement.
- **Planning:** Identifying environmental aspects, setting objectives and targets, and developing plans to achieve them.
- **Implementation:** Assigning roles, providing training, establishing procedures, and managing documentation.
- **Checking and corrective action:** Monitoring, measuring, auditing, and taking corrective action.
- **Management review:** Regular review of the EMS by top management to ensure effectiveness.

**Benefits of ISO 14001 implementation:**

- **Regulatory compliance:** Ensures compliance with environmental laws and regulations.
- **Cost savings:** Reduces waste, energy, and resource costs.
- **Competitive advantage:** Certification is often a requirement for tenders and contracts.
- **Risk management:** Identifies and mitigates environmental risks.
- **Reputation:** Demonstrates environmental responsibility to stakeholders.
- **Continuous improvement:** Provides a framework for ongoing environmental performance improvement.
- **Employee engagement:** Involving employees in environmental management fosters a culture of sustainability.

> 🛡️ **Note:** ISO 14001 certification is voluntary but highly valued in industry — it demonstrates a commitment to environmental responsibility.

> **Malayalam support:** ISO 14000 □□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□ □□□□ □□□□□□□□□□□□. ISO 14001 □□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□ □□□□□□□□□□□□□.

## ▸Biotechnology and Environmental Protection

**Biotechnology** is the use of living organisms (microorganisms, plants, enzymes) or their systems to develop products and processes for specific uses.

**Applications of biotechnology in environmental protection:**

- **Bioremediation:** Using microorganisms to degrade or detoxify pollutants in soil and water. Examples: Oil spill cleanup (oil-degrading bacteria), heavy metal removal (bioaccumulation).
- **Phytoremediation:** Using plants to absorb, accumulate, or degrade pollutants. Examples: Sunflowers for heavy metals, water hyacinth for wastewater treatment.
- **Biofertilizers:** Using microorganisms (nitrogen-fixing bacteria, mycorrhizal fungi) to improve soil fertility and reduce chemical fertilizer use.
- **Biopesticides:** Using natural organisms or substances (Bacillus thuringiensis, neem) for pest control, reducing chemical pesticide use.
- **Biofuels:** Using biomass (agricultural waste, algae) to produce renewable fuels (bioethanol, biodiesel).
- **Biosensors:** Using biological elements (enzymes, antibodies) to detect environmental pollutants.

> 💡 **Note:** Biotechnology offers nature-based solutions to environmental challenges — they are often more sustainable and less harmful than chemical-based approaches.

> **Malayalam support:** [illegible] (Biotechnology) [illegible] [illegible], [illegible], [illegible], [illegible], [illegible].

### ▸Green Chemistry for Sustainable Development

**Green chemistry** is the design of chemical products and processes that reduce or eliminate the use and generation of hazardous substances. It is based on 12 principles that guide sustainable chemical design.

**Twelve Principles of Green Chemistry:**

1. **Prevent waste:** Design processes that minimize waste generation.
2. **Atom economy:** Maximize the incorporation of all materials used in the process into the final product.
3. **Less hazardous chemical synthesis:** Design synthesis that uses and produces substances with minimal toxicity.
4. **Designing safer chemicals:** Design chemicals that are effective yet non-toxic.
5. **Safer solvents and auxiliaries:** Avoid hazardous solvents and use safer alternatives.
6. **Design for energy efficiency:** Conduct reactions at ambient temperature and pressure.
7. **Use of renewable feedstocks:** Use renewable resources (biomass) instead of fossil resources.
8. **Reduce derivatives:** Minimize the use of protecting groups and other unnecessary steps.
9. **Catalysis:** Use catalytic reagents (small amounts) instead of stoichiometric reagents (large amounts).
10. **Design for degradation:** Design chemicals that degrade into harmless products.
11. **Real-time analysis for pollution prevention:** Monitor chemical processes in real-time to prevent pollution.
12. **Inherently safer chemistry for accident prevention:** Design processes that minimize the risk of accidents.

> 🛡️ **Key insight:** Green chemistry fundamentally changes the way we design and manufacture chemicals — moving from "end-of-pipe" pollution control to "source reduction."

> **Malayalam support:** [illegible] [illegible] (Green chemistry) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. 12 [illegible] [illegible].

> 📌 **Module IV Summary:** This module covered sustainable engineering principles, zero waste concepts (3R principles, case studies), energy recovery from waste (biochemical and thermochemical methods), carbon credit and carbon footprint, environmental management in fabrication, ISO 14000, biotechnology for environmental protection, and the twelve principles of green chemistry for sustainable development.

QUICK REFERENCE

# Formula Bank

### Carbon Footprint

$$CF = \Sigma\,(AD \times EF)$$

AD = Activity Data, EF = Emission Factor ($CO_2e$ per unit)

### Energy from Waste (Incineration)

$$E = \eta \times HHV \times m$$

$\eta$ = efficiency, HHV = Higher Heating Value, m = mass of waste

### Carbon Credit Value

$$\text{Value} = \text{Credits} \times \text{Price}$$

1 credit = 1 tonne $CO_2$ equivalent

### Efficiency of Energy Recovery

$$\eta = (\text{Energy Output} / \text{Energy Input}) \times 100\%$$

Measured as percentage

VISUALS

# Diagram Library for Rapid Revision

### 🌍 Environment Segments

Atmosphere, Lithosphere, Hydrosphere, Biosphere

🌐

### 🕸 Food Web

Interconnected energy flow in ecosystems

🕸

### 🌱 Sustainability Pillars

Environmental, Social, Economic balance

♻️

### ♻️ Zero Waste Hierarchy

Refuse → Reduce → Reuse → Recycle → Recover

🗑️

SELF-LEARNING

## Student Activities

Official self-learning activities from the syllabus — recommended for independent study and project work.

**WEEK 1 · MODULE I**

### 🌿 Case Study on Unsustainable Energy Projects

Present a case study on unsustainable energy projects that affect surrounding landscapes or ecosystems.

⏱️ 4.5 hours

**WEEK 2 · MODULE I**

### 🌍 Report on Local Environment Issues

Write a report on local environment issues regarding air or water pollution.

⏱️ 4.5 hours

**WEEK 3 · MODULE I**

### 📊 Chart on Energy Transfer in Ecosystem

Prepare a chart showing energy transfer in an ecosystem.

⏱️ 4.5 hours

**WEEK 4 · MODULE I**

### ⚖️ Report on Environmental Protection Laws

Analyze environmental protection laws in India and submit a report on it.

⏱️ 4.5 hours

**WEEK 5 · MODULE II**

### 📝 Note on Ethical Theories

Write a note on ethical theories based on specific relevant cases.

⏱️ 4.5 hours

WEEK 6 · MODULE II

### 🔗 Ethical Principles — Environmental Practices Diagram

Create diagrams showing connections between ethical principles and environmental practices.

⏱️ 4.5 hours

WEEK 7 · MODULE II

### 🎨 Posters on Environmental Ethics

Make posters expressing their ethical stance on environmental issues.

⏱️ 4.5 hours

WEEK 8 · MODULE II

### 📖 Key Ethical Approaches

Describe the key ethical approaches to the environment.

⏱️ 4.5 hours

WEEK 9 · MODULE III

### ☀️ Renewable Energy Report

Submit a report on successful implementation of any renewable energy sources in India.

⏱️ 4.5 hours

WEEK 10 · MODULE III

### 🗑️ Campus Waste Audit

Prepare a waste audit report of the campus including suggestions for zero waste approach.

⏱️ 4.5 hours

WEEK 11 · MODULE III

### 🌴 Life Cycle Analysis of Kerala Product

Write a life cycle analysis report of a common product used in Kerala (e.g., coconut, bamboo, rubber-based product) and present findings on its sustainability.

⏱️ 4.5 hours

WEEK 12 · MODULE III

### 🔋 Energy Consumption Analysis

Analyze the energy consumption pattern of the campus and propose sustainable alternatives to reduce consumption.

⏱️ 4.5 hours

WEEK 13 · MODULE IV

### ♻️ Plastic Waste Management Case Study

Undertake a case study on plastic waste management in Kerala and prepare a report.

⏱️ 4.5 hours

WEEK 14 · MODULE IV

### ⚡ Energy Recovery Methods Report

Write a report on different types of energy recovery methods.

⏱️ 4.5 hours

WEEK 15 · MODULE IV

### 🧪 Green Chemistry Applications

Write a note on the applications of green chemistry.

⏱️ 4.5 hours

**Total Self-Learning Hours:** 67.5 hours (official syllabus)

EXAMINATION

# Assessment Methodology

**Examination Scheme**

| Component | Marks | Details |
|---|---|---|
| **CIE (Continuous Internal Evaluation)** | 40 | Based on internal assessments, assignments, and class performance. |
| **ESE (End Semester Examination)** | 60 | Written examination at the end of the semester. |
| **Total** | **100** | |

## Series Test Pattern

- Two series tests are typically conducted during the semester.
- Each series test covers approximately two modules.
- Questions are designed to test understanding and application.

## ESE (End Semester Examination) Pattern

- **Part A:** Short answer questions (2 marks each) — covering all modules.
- **Part B:** Medium answer questions (5 marks each) — covering all modules.
- **Part C:** Long answer questions (10 marks each) — covering all modules.
- Total marks: 60

💡 **Exam tip:** Focus on understanding concepts, key definitions, and the ability to explain with examples. Draw diagrams where applicable to show understanding.

PRACTICE

# Module-wise Questions with Answers

## Module I — Ecosystem & Environment

**Q: Define environment and explain its four segments.**

**A:**
**Environment** is the sum total of all external conditions and influences that affect the life, development, and survival of an organism.
**Four segments:**
- **Atmosphere:** Gaseous envelope — $N_2$ (78%), $O_2$ (21%), trace gases.
- **Lithosphere:** Solid outer layer — rocks, minerals, soil.
- **Hydrosphere:** All water bodies — oceans, rivers, lakes, groundwater.
- **Biosphere:** The zone of life — all ecosystems.

**Q: Explain the components of an ecosystem with examples.**

**A:**
**Biotic (Living) Components:**
- **Producers:** Green plants, algae (photosynthesis).
- **Consumers:** Herbivores (primary), carnivores (secondary), top carnivores (tertiary).
- **Decomposers:** Bacteria, fungi (break down organic matter).

**Abiotic (Non-living) Components:**
- **Physical:** Sunlight, temperature, rainfall, humidity, soil.
- **Chemical:** Soil pH, mineral content, dissolved oxygen, nutrient availability.

**Q: What is the difference between a food chain and a food web?**

**A:**
- **Food chain:** Single, linear path of energy flow (e.g., grass → deer → lion).
- **Food web:** Multiple interconnected food chains showing complex feeding relationships.
- **Key difference:** A food web is a more realistic representation of an ecosystem's energy flow.

**Q: Briefly discuss the Bhopal gas tragedy and its environmental lessons.**

**A:**
**Bhopal Gas Tragedy (1984):** Methyl isocyanate (MIC) gas leaked from the Union Carbide pesticide plant in Bhopal, India. Over 3,800 died immediately and tens of thousands suffered chronic health effects.
**Lessons:**
- Industrial safety regulations must be strictly enforced.

- Community emergency preparedness is critical.
- Corporate accountability is essential.

**Q: List five major international environmental conventions and their key focus areas.**

**A:**

- **Stockholm Conference (1972):** First UN conference on human environment.
- **Montreal Protocol (1987):** Phases out ozone-depleting substances.
- **Kyoto Protocol (1997):** Legally binding emissions targets for developed countries.
- **Paris Agreement (2015):** Global framework to limit warming to below 2°C.
- **Basel Convention (1989):** Controls transboundary movement of hazardous waste.

**Q: What are the main provisions of the Environment (Protection) Act, 1986?**

**A:**

- Umbrella legislation for environmental protection in India.
- Empowers the Central Government to set standards for emissions and effluents.
- Requires Environmental Impact Assessment (EIA) for certain projects.
- Provides penalties for non-compliance (imprisonment up to 5 years and fines).

## Module II — Ethics — Professional Ethics — Environmental Ethics

**Q: Define ethics and explain its three main types.**

**A:**

**Ethics:** The branch of philosophy that deals with moral principles and rules of conduct.

- **Normative Ethics:** Prescribes how people ought to behave.
- **Meta-Ethics:** Examines the nature of moral concepts and language.
- **Applied Ethics:** Applies ethical principles to specific real-world situations.

**Q: Compare deontology and consequentialism with examples.**

**A:**

- **Deontology:** Actions are right if they follow universal moral rules/duties. (Example: Always tell the truth, regardless of consequences.)
- **Consequentialism:** Actions are judged by their outcomes. (Example: A lie is acceptable if it saves a life.)
- **Key difference:** Deontology focuses on duties; consequentialism focuses on results.

**Q: Distinguish between personal ethics and professional ethics.**

**A:**

- **Personal ethics:** Based on individual moral values, upbringing, religion — applied in personal life.
- **Professional ethics:** Based on professional codes and standards — applied in the workplace.

- Professional ethics often provides a more formalized set of expectations, while personal ethics is more subjective.

**Q: What is environmental ethics? Explain its importance.**

**A:**

**Environmental ethics** extends moral consideration to the natural environment — non-human animals, plants, ecosystems, and the planet.

**Importance:**

- Provides a moral framework for environmental protection.
- Recognizes the intrinsic value of nature.
- Promotes intergenerational justice.
- Guides sustainable development and conservation.

**Q: Differentiate between anthropocentrism, biocentrism, and ecocentrism.**

**A:**

- **Anthropocentrism:** Human-centered — nature has value only for human use.
- **Biocentrism:** Life-centered — all living beings have intrinsic value.
- **Ecocentrism:** Ecosystem-centered — the whole ecosystem has intrinsic value.

## Module III — Sustainable Development

**Q: Define sustainability and explain its three pillars.**

**A:**

**Sustainability:** Meeting the needs of the present without compromising the ability of future generations to meet their own needs.

- **Environmental:** Protecting natural resources and ecosystems.
- **Social:** Ensuring equity, justice, and community well-being.
- **Economic:** Creating viable and equitable economic systems.

**Q: List four major anthropogenic effects on the environment.**

**A:**

- **Pollution:** Air, water, soil, noise.
- **Deforestation:** Loss of forests for agriculture and urbanization.
- **Soil erosion:** Loss of topsoil due to unsustainable land use.
- **Mass extinction:** Loss of biodiversity due to human activities.

**Q: Explain the concept of renewable energy and give examples.**

**A:**

**Renewable energy** is energy from natural sources that are replenished at a rate equal to or faster than they are consumed.

**Examples:**

- Solar (photovoltaic and thermal).
- Wind (onshore and offshore turbines).

- Hydro (dams and run-of-river).
- Biomass (combustion and gasification).
- Geothermal (heat from the Earth's interior).

**Q: What is Environmental Impact Assessment (EIA)? Explain its key steps.**

**A:**
**EIA** is a systematic process for evaluating the environmental consequences of a proposed project.
**Key steps:**
- Screening → Scoping → Impact analysis → Mitigation → Environmental Management Plan → Public consultation → Decision-making → Monitoring.

**Q: Describe the environmental impacts of urbanization and sustainable urban planning strategies.**

**A:**
**Impacts:** Air and water pollution, heat island effect, biodiversity loss, resource consumption, waste generation.
**Sustainable strategies:**
- Green buildings, urban parks, sustainable transportation.
- Integrated water management, waste management, compact city design.

## Module IV — Ethical Engineering and Environmental Management

**Q: What is sustainable engineering and what is the triple bottom line?**

**A:**
**Sustainable engineering** is the design and operation of systems that meet human needs while preserving the environment for future generations.
**Triple Bottom Line:**
- **Economic:** Projects must be viable and profitable.
- **Social:** Projects must benefit communities and ensure equity.
- **Environmental:** Projects must protect and enhance the environment.

**Q: Explain the 3R principle and its importance in waste management.**

**A:**
- **Reduce:** Using fewer resources and generating less waste.
- **Reuse:** Using items multiple times instead of disposing after single use.
- **Recycle:** Processing waste materials into new products.

**Importance:** Conserves resources, reduces energy consumption, reduces greenhouse gas emissions, reduces landfill use, creates economic opportunities.

**Q: Compare biochemical and thermochemical methods of energy recovery from waste.**

**A:**

- **Biochemical:** Landfill gas recovery (methane capture), composting (aerobic decomposition).
- **Thermochemical:** Incineration (burning), gasification (syngas production), pyrolysis (thermal decomposition in absence of oxygen).
- **Key difference:** Biochemical uses biological processes; thermochemical uses heat and chemical reactions.

**Q: What is a carbon credit and how does carbon trading work?**

**A:**

**Carbon credit:** A tradable permit representing one metric ton of $CO_2$ equivalent that has been reduced or removed from the atmosphere.
**Carbon trading:** Entities with emissions caps can buy/sell credits — those that reduce emissions below their cap can sell credits to those that exceed their cap. This creates a market incentive to reduce emissions.

**Q: What is ISO 14001 and what are its key benefits?**

**A:**

**ISO 14001** is the international standard for Environmental Management Systems (EMS).
**Benefits:**

- Regulatory compliance, cost savings, competitive advantage.
- Risk management, reputation enhancement, continuous improvement.
- Employee engagement in sustainability.

**Q: List the twelve principles of green chemistry.**

**A:**

1. Prevent waste
2. Atom economy
3. Less hazardous synthesis
4. Designing safer chemicals
5. Safer solvents
6. Energy efficiency
7. Renewable feedstocks
8. Reduce derivatives
9. Catalysis
10. Design for degradation
11. Real-time analysis
12. Inherently safer chemistry

PRACTICE

## Model Question Paper

Based on the official pattern — this is a practice model paper, not an official SITTTR model question paper.

## Course 2005 — Environmental Sustainability and Professional Ethics

**Time:** 2½ Hours | **Total Marks:** 60

**Part A — Short Answer Questions (2 marks each, Answer any 5)**

1. Define environment and list its four segments.
2. What is a food chain? Give one example.
3. What is the Montreal Protocol?
4. Distinguish between personal ethics and professional ethics.
5. What is the triple bottom line?
6. Define carbon footprint.
7. What is the 3R principle in waste management?

**Part B — Medium Answer Questions (5 marks each, Answer any 4)**

1. Explain the components of an ecosystem with suitable examples.
2. Discuss the Bhopal gas tragedy and its lessons for industrial safety.
3. Explain the three ethical approaches to the environment — anthropocentrism, biocentrism, and ecocentrism.
4. What is sustainable development? Explain its strategies.
5. Describe the Environmental Impact Assessment (EIA) process.
6. Explain zero waste concepts and practices with a case study.

**Part C — Long Answer Questions (10 marks each, Answer any 2)**

1. Explain energy flow in an ecosystem with the help of a food chain and food web. Add a note on the importance of understanding these concepts for environmental management.
2. Discuss the major international conventions and conferences on environmental protection. What is their significance in global environmental governance?
3. Explain the concept of sustainable engineering, the triple bottom line, and life cycle analysis. How do these concepts contribute to sustainable development?
4. What is green chemistry? Explain the twelve principles of green chemistry with examples. How does green chemistry contribute to environmental sustainability?

QUICK RECAP

# Revision Cards

### 📘 Module I — Ecosystem & Environment

- Environment: Atmosphere, Lithosphere, Hydrosphere, Biosphere
- Ecosystem = Biotic + Abiotic components
- Food chain (linear) vs Food web (network)
- Pollution: global, regional, local; persistent, non-persistent
- Key conventions: Montreal, Kyoto, Paris, Basel
- Indian laws: Forest Act 1927, Water Act 1974, Air Act 1981, EPA 1986

### ⚖️ Module II — Ethics

- Ethics: normative, meta, applied

- Theories: deontology (duty), consequentialism (outcome), virtue ethics (character)
- Personal vs Professional ethics
- Environmental ethics: intrinsic value of nature
- Approaches: anthropocentrism, biocentrism, ecocentrism

### 🌱 Module III — Sustainable Development

- Sustainability = Environment + Social + Economic
- Anthropogenic impacts: pollution, deforestation, erosion, extinction
- Renewable energy: solar, wind, hydro, biomass
- EIA: screening → scoping → impact → mitigation → EMP → monitoring
- India initiatives: Smart City, Swachh Bharat, National Water Mission
- Sustainable urban planning: green buildings, parks, transit

### ♻️ Module IV — Ethical Engineering

- Sustainable engineering: LCA, triple bottom line
- Zero waste: Refuse, Reduce, Reuse, Recycle, Rot, Rethink
- Energy recovery: biochemical (landfill, compost) & thermochemical (incineration, gasification, pyrolysis)
- Carbon credit (tradable permit) & Carbon footprint ($CO_2e$ emissions)
- ISO 14001: Environmental Management System
- Green chemistry: 12 principles — safer, cleaner, sustainable

## ⚠️ Common Mistakes to Avoid

- **Confusing food chain and food web:** Remember — food chain is linear; food web is a network.
- **Mixing up ethical theories:** Deontology = duty, Consequentialism = outcome, Virtue ethics = character.
- **Confusing anthropocentrism vs biocentrism vs ecocentrism:** Anthropo = human; bio = life; eco = ecosystem.
- **Mixing up EIA and Environmental Audit:** EIA is prospective (before project); audit is retrospective (during/after).
- **Forgetting the 3R order:** Reduce → Reuse → Recycle (priority order).
- **Confusing carbon credit and carbon footprint:** Credit = permit to emit; Footprint = total emissions.

## 📋 Exam Checklist

- ✅ Module I concepts clear
- ✅ Module II ethical frameworks
- ✅ Module III sustainability principles
- ✅ Module IV zero waste & green chemistry
- ✅ Formulas (carbon footprint, efficiency)
- ✅ Diagrams (food web, environment segments)
- ✅ Case studies (Bhopal, Minamata, Chernobyl)
- ✅ Conventions (Montreal, Kyoto, Paris)
- ✅ Indian laws (EPA 1986, Water Act, Air Act)
- ✅ Practice questions completed

TERMS

# Glossary

**Important Terms and Definitions**

| Term | Definition |
|---|---|
| **Ecosystem** | A community of living organisms interacting with their non-living environment. |
| **Food Chain** | A linear sequence of organisms through which energy and nutrients pass. |
| **Food Web** | A network of interconnected food chains. |
| **Pollutant** | A substance or energy that causes pollution. |
| **Anthropocentrism** | Human-centered ethics — nature has value only for human use. |
| **Biocentrism** | Life-centered ethics — all living beings have intrinsic value. |
| **Ecocentrism** | Ecosystem-centered ethics — ecosystems as wholes have intrinsic value. |
| **Sustainability** | Meeting present needs without compromising future generations. |
| **Triple Bottom Line** | Economic, social, and environmental sustainability. |
| **Zero Waste** | A philosophy that aims to eliminate waste through circular design. |
| **Carbon Credit** | A tradable permit for one tonne of $CO_2$ equivalent. |
| **Carbon Footprint** | Total greenhouse gas emissions of an entity or activity. |
| **ISO 14001** | International standard for Environmental Management Systems. |
| **Green Chemistry** | Design of chemical processes that reduce hazardous substances. |
| **EIA** | Environmental Impact Assessment — evaluating project consequences. |
| **LCA** | Life Cycle Analysis — assessing environmental impacts over a product's life. |

RESOURCES

# References

## Textbooks


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### Web Resources (officially listed)

- — Modules I & IV
- — Modulrtal.org.in" target="_blank" rel="noopener">www.indiaenvironmentportal.org.in — Modules I & IV
- <
- 